ELKIK'S LEMMA: THE SINGULAR IDEAL IS THE RADICAL OF THE IDEAL GENERATED BY STRICTLY OR ELEMENTARY STANDARD ELEMENTS

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Editorial note. Input lemma-parse-equation-strictly-standard-one is same-chapter NOT carded (dropped in judgment as a pure pointer to Algebra matrix-left-inverse) and is declared external.

Allowed standard prerequisites (editorial).

- Strictly standard implies smooth after inverting the element. lemma-parse-equation-strictly-standard-one (same chapter, NOT carded, declared external). Turns the two strict-standard equations into a left inverse of the conormal differential.
- Localization of the naive cotangent complex. Algebra, lemma localize-NL (cross-chapter, named). Concludes smoothness of R -> A_a from the NL criterion.
- Finding a strictly standard element at a smooth point. lemma-find-strictly-standard (carded L26794). Supplies the reverse inclusion H_{A/R} contained in H_e.

Source reference. Original reference 00S7: focused-seeds/geometry/S9-smo-elkik/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original definition 07C7: sources/stacks-smoothing/pinned-original-source.tex, source lines 409–409. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 07ER: sources/stacks-smoothing/pinned-original-source.tex, source lines 429–429. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 07ES: sources/stacks-smoothing/pinned-original-source.tex, source lines 435–435. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:smoothing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 07C6: sources/stacks-smoothing/pinned-original-source.tex, source lines 359–359. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07ET: sources/stacks-smoothing/pinned-original-source.tex, source lines 446–446. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1 (Elkik)**.** *Let $R \to A$ be a ring map of finite presentation. The singular ideal $H_{A/R}$ is the radical of the ideal generated by strictly standard elements in A over R and also the radical of the ideal generated by elementary standard elements in A over R.*

Proof. Assume a is strictly standard in A over R. We claim that A_a is smooth over R, which proves that $a \in H_{A/R}$. Namely, let $A = R[x_1, \ldots, x_n]/(f_1, \ldots, f_m)$, c, and $a' \in A$ be as in Definition 07C7. Write $I = (f_1, \ldots, f_m)$ so that the naive cotangent complex of A over R is given by $I/I^2 \to \bigoplus A\mathrm{d}x_i$. Assumption (07ES) implies that $(I/I^2)_a$ is generated by the classes of $f_1, \ldots, f_c$. Assumption (07ER) implies that the differential $(I/I^2)_a \to \bigoplus A_a\mathrm{d}x_i$ has a left inverse, see Lemma 07ET. Hence $R \to A_a$ is smooth by definition and Algebra, Lemma 00S7.

Let $H_e, H_s \subset A$ be the radical of the ideal generated by elementary, resp. strictly standard elements of A over R. By definition and what we just proved we have $H_e \subset H_s \subset H_{A/R}$. The inclusion $H_{A/R} \subset H_e$ follows from Lemma 07C6. $\square$